

TRI-OCTAGON MATHEMATICAL RESEARCH

Measurement Geometry, Perspective, and Vesica Interfaces in the Tri-Octagon Model

Hilmir Frímann Halldórsson

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Abstract

We give an exact account of the measurement geometry and adopted lens-to-state interface of the Tri-Octagon model. The width-one folded module is enclosed by the minimum centered vertical cylinder of radius $1/\sqrt{3}$ and half-height $1/2$. Its highest measurement ring is distinct from the nonplanar upper rim. Explicit projections separate normal, seam and tangent directions, and distinguish a measurement polygon from a regular reference hexagon. Historical changes of units are separated from fixed-centre changes of construction. For fixed nonzero extraction data, the lens initializer retains only the ratio d/r ; its fibres are positive scaling rays. Ratio recovery from amplitude is sharply Lipschitz, while angle recovery at tangency is Hölder. Deterministic vector error and finite-transfer attenuation explain conditioning without restoring absent absolute scale. Conditional enclosing hosts are retained in appendices. The cylinder is a static measurement definition and the oriented lens remains a kinematic interface; neither is assigned a material or rotational law.

Keywords: folded geometry; measurement scaffold; lens identifiability; inverse conditioning; SRG preparation

Principal results and hypotheses. Lengths use the current octagon width as one. With the fixed centre and vertical axis, $R_{\min} = 1/\sqrt{3}$, $H_{\min} = 1/2$ (Section 3). For known fixed $v \neq 0$, $\Omega_0 = Gv$ determines $q = d/r$, but not r ; each fibre is $\{(r, qr) : r > 0\}$ (Section 6). The inverse obeys $|Q(t_1) - Q(t_2)| \leq \pi|t_1 - t_2|$, whereas $\theta(t) \sim (3\pi/4)^{1/3}t^{2/3}$ at tangency (Section 7). If $v = 0$, the whole lens domain collapses.

Notation. S : folded surface; o : centre; R, H : cylinder radius and half-height; r, d : lens radius and separation; $q = d/r$; $t = G$: amplitude gain; v : fixed extracted vector. Appendix C gives the full crosswalk. Master labels M1–M38 and MA1–MA9 are retained.

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1 The picture and the question

The present construction consists of a fixed, folded polygonal surface and a surrounding coordinate scaffold. The owner's TL4 clarification gives the scaffold a precise purpose: the cylinder and its highest circular ring are **measurement geometry**, enclosing the centre and the entire Tri-Octagon. They have no specified material response or evolution. A separate Vesica-derived eye may eventually be oriented relative to this frame, but no rotation law has been supplied.

That clarification changes the mathematical question. An enclosing reference cylinder can be specified from exact coordinate extrema; selecting a material host would additionally require a choice of surface, attachment, and physical law. Earlier TL3 calculations explored conditional hosts. Their containment formulas remain valid, but none is needed to define the measurement cylinder, and none was adopted as material physics.

The resulting account has two parts. First, which lengths, directions, and projected outlines are actually supplied by the folded module? Second, which lens information reaches the canonical initial state through the existing adopted initializer? The first determines a reference frame. The second is an identifiability and conditioning problem. Neither supplies dynamics for a surrounding eye.

Definition/adoption and source boundary. TL4's meaning of "top circle" is a current owner clarification. TL0's earlier historical recovery remains **TOP_CIRCLE_DISTINCT_OBJECT = OPEN / NOT_FORMALIZED**: the clarification does not retrospectively identify every historical ring or drawing. No NASA attribution, missing DMQPF explanation, or physical interpretation is inferred from recollection. [TL0 §§1–9; TL4 "Scope and evidence."]

2 The exact width-one folded module

All lengths below use the current flat-to-flat octagon width as one unit. This is a geometric normalization, not a calibrated physical length. Put

$$s = \sqrt{2} - 1, \quad a = \frac{1}{2}, \quad p = \frac{1}{2\sqrt{3}}, \quad o = (0, \sqrt{3}/6, 0), \quad e_z = (0, 0, 1). \quad (\text{M1})$$

Here s is an octagon side, a its apothem and vertical half-height, p the distance from the symmetry centre o to a face centre. These are different lengths. The local filled octagon has vertices

$$\begin{aligned} &(-1/2, -s/2), (-s/2, -1/2), (s/2, -1/2), (1/2, -s/2), \\ &(1/2, s/2), (s/2, 1/2), (-s/2, 1/2), (-1/2, s/2). \end{aligned} \quad (\text{M2})$$

In the source's material coordinates (u, z) , its three affine panel maps at $\beta = \pi/3$ are

$$\begin{aligned} P_1(u, z) &= (-1/2 + (1/2 - u) \cos \beta, (1/2 - u) \sin \beta, z), \\ P_2(u, z) &= (u, 0, z), \\ P_3(u, z) &= (1/2 - (1/2 + u) \cos \beta, (1/2 + u) \sin \beta, z). \end{aligned} \quad (\text{M3})$$

Their union S is the union of the **filled panels**, with 18 welded vertices, three seams and two nonplanar nine-edge rims. There are no top or bottom caps. Twenty-four face-vertex incidences do not mean twenty-four different vertices.

For measurement, the centred face description is simpler. With face labels $A = P_1, B = P_2, C = P_3$, define

$$n_A = (-\sqrt{3}/2, 1/2, 0), \quad n_B = (0, -1, 0), \quad n_C = (\sqrt{3}/2, 1/2, 0), \quad t_i = e_z \times n_i. \quad (\text{M4})$$

Each face centre is $c_i = o + pn_i$. As a point set, its panel is exactly

$$x - o = pn_i + ut_i + ze_z, \quad |u| \leq \frac{1}{2}, \quad |z| \leq U(u) := \min\left(\frac{1}{2}, \frac{1}{\sqrt{2}} - |u|\right). \quad (\text{M5})$$

The material sign of u in (M3) need not agree on every face with the tangent-oriented u in (M5); the polygonal sets agree.

Derived identity. Write $\rho = \|(I - e_z e_z^T)(x - o)\|$. Orthogonality of the frame gives

$$\rho^2 = p^2 + u^2, \quad p \leq \rho \leq \frac{1}{\sqrt{3}}, \quad |z| \leq \frac{1}{2}. \quad (\text{M6})$$

Every vertex has the same centred squared norm

$$B^2 = p^2 + \frac{1}{4} + \frac{s^2}{4} = \frac{13}{12} - \frac{\sqrt{2}}{2}. \quad (\text{M7})$$

The high vertices have $(\rho, |z|) = (v_6, 1/2)$, where $v_6 = \sqrt{5/6 - \sqrt{2}/2} \simeq 0.355283763$; seam vertices have $(\rho, |z|) = (1/\sqrt{3}, s/2)$. The largest horizontal radius occurs below the top plane. A circle fitted only to the top vertices would miss the seams. [Inherited geometry: native geometry.py and face_state.py; TL3 §2, (1)–(3); TL4 “Sharp enclosure.”]

3 Measurement cylinder and highest ring

Conditional measurement definition. Fix the centre o and vertical axis $o + \mathbb{R}e_z$. For $R, H > 0$, let

$$\boxed{C(R, H) = \{x : \rho \leq R, \quad |z| \leq H\}}, \quad z = (x - o) \cdot e_z. \quad (\text{M8})$$

Here H is **half-height**; the full height is $2H$. Set $e_x = (1, 0, 0) = t_B$, $e_y = (0, 1, 0)$. Its highest circular ring is

$$\boxed{\Gamma_+(R, H) = \{o + R(\cos \psi e_x + \sin \psi e_y) + He_z : 0 \leq \psi < 2\pi\}}. \quad (\text{M9})$$

The solid cylinder’s full maximum-height set is a disk. The owner’s top ring is that disk’s circular boundary, not the entire disk, the lateral surface, or the vertical axis.

Exact theorem, fixed registration. The complete panel surface satisfies

$$\boxed{S \subset C(R, H) \iff R \geq R_{\min} = \frac{1}{\sqrt{3}}, \quad H \geq H_{\min} = \frac{1}{2}}. \quad (\text{M10})$$

To prove sufficiency, apply (M6) to every panel point. Necessity follows because $|u| = 1/2$ is attained on seams and $|z| = 1/2$ on short horizontal edges. These independent extrema require both inequalities. The centre already lies in every such cylinder. Convexity also gives enclosure of $\text{conv}(S \cup \{o\})$, without declaring that convex hull a material core.

Thus the minimum radius is $0.577350269\dots$, diameter $2/\sqrt{3}$, half-height $1/2$, and full height 1. These minima are for the stipulated centred vertical cylinder; no optimization over tilted or displaced cylinders is asserted.

At equality the lateral surface touches the seams for $|z| \leq s/2$, while the top and bottom disks meet the short horizontal edges. The native shell does **not** touch the highest circular ring: at $z = 1/2$ its radius is at most $v_6 < 1/\sqrt{3}$. Its upper rim contains six vertices at height $1/2$ and three at $s/2$. Three noncollinear high vertices fix the plane $z = 1/2$, excluding the other three, so that rim cannot be a Euclidean circle. A topological boundary cycle and a measurement circle are different objects. [TL0 §6; TL4 (7)–(10), written sharpness proof.]

For an additional centred horizontal measurement polygon with largest vertex radius ρ_P , placed at height z_0 , the joint enclosure is exactly

$$R \geq \max(1/\sqrt{3}, \rho_P), \quad H \geq \max(1/2, |z_0|). \quad (\text{M11})$$

A planar footprint alone supplies no nonzero height. This applies separately to a chosen regular reference hexagon; it does not turn that hexagon into a shell section.

Static folded module and measurement cylinder

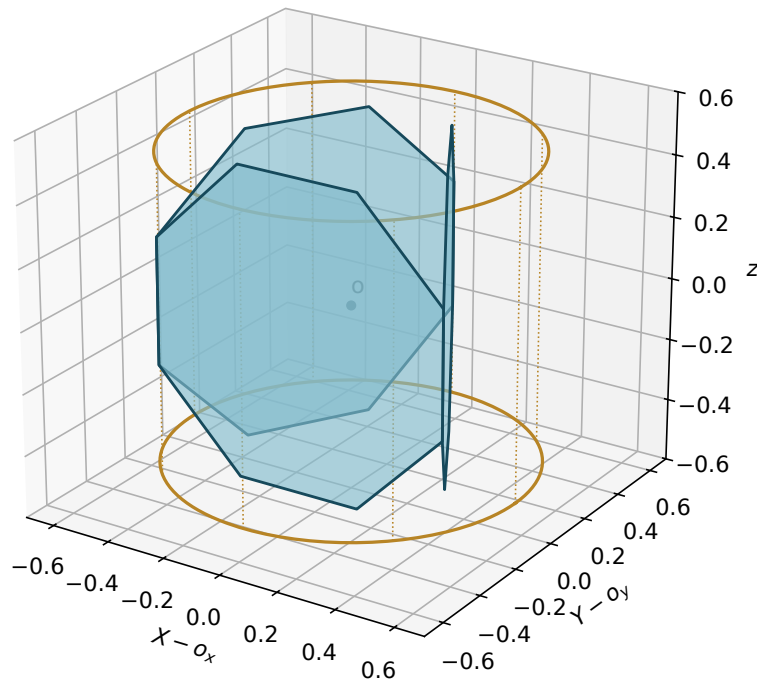


Figure 1: Exact coordinate construction in width-one units centered at o : three filled panels (blue) and added static cylinder rings/guides (gold), with $R = 1/\sqrt{3}$, $H = 1/2$. Cylinder guides are measurement geometry, not shell edges. Equations (M1)–(M11); formula-based rendering, no dynamics.

4 Projections, directions, and the two hexagons

4.1 What a top view actually shows

Top projection is $\pi_z(x) = (e_x \cdot (x - o), e_y \cdot (x - o))$. In (M5) it removes z , leaving three complete segments $pn_i + ut_i$, $-1/2 \leq u \leq 1/2$. Hence

$$\pi_z(S) = \text{the perimeter of an equilateral triangle of side 1.} \quad (\text{M12})$$

It is not a filled triangle or hexagon. The filled triangle is its convex hull. The highest subset projects to three short segments of length s . Connecting their six endpoints by three additional measurement chords produces an alternating hexagon. Those extra chords are not edges of the native shell.

For comparison, a side view along the line n_B , with screen (t_B, e_z) , retains (x, z) . Let $U_z = \min(1/2, 1/\sqrt{2} - |z|)$. The panels project at fixed z to

$$[-U_z, U_z], \quad [-1/4 - U_z/2, -1/4 + U_z/2], \quad [1/4 - U_z/2, 1/4 + U_z/2].$$

They overlap because $U_z \geq s/2 > 1/6$. Their union proves

$$\pi_{t_B, e_z}(S) = \{(x, z) : |z| \leq 1/2, |x| \leq 1/4 + U_z/2\}. \quad (\text{M13})$$

This is a filled nonregular octagon, with corners $(\pm 1/2, \pm s/2)$ and $(\pm 1/(2\sqrt{2}), \pm 1/2)$. A triangle perimeter in one view and a filled octagon in another are consistent projections of exactly the same surface. [TL4 (12)–(18).]

4.2 Directed rays are not unoriented axes

Azimuths below run from $+x = t_B$ towards $+y$. A ray distinguishes v from $-v$; the line $\mathbb{R}v$ does not.

Object	Directed azimuths	Unoriented lines modulo 180°
Face normals and face-centre rays	30, 150, 270°	30, 90, 150°
Rays to projected seams; gap bisectors	90, 210, 330°	Same normal lines
Tangents t_i	0, 120, 240°	0, 60, 120°
CCW connector directions	60, 180, 300°	Same tangent lines
Regular aligned hexagon vertex rays	0, 60, 120, 180, 240, 300°	Same tangent lines

The triangle corner opposite face i is $-2pn_i$. Thus the seam-position rays are exactly $-n_i$. The seam itself is a vertical segment: its spatial tangent is e_z , not the horizontal ray to its projected point. Similarly $\{R_{60}t_i\} = \{-t_i\}$. The normal and tangent line sets differ by 30° as unordered triples and are inequivalent under native D_{3h} , although both are derived from the same frame. Three distinguished lines in a plane do not mean three independent coordinate dimensions.

For the aligned endpoint construction write

$$A_i = pn_i - \frac{s}{2}t_i, \quad B_i = pn_i + \frac{s}{2}t_i, \quad g_{\text{gap}} = \sqrt{3}p - \frac{s}{2}. \quad (\text{M14})$$

The selected edges have length s ; direct subtraction of adjacent endpoints gives connectors of length g_{gap} in directions $R_{60}t_i$. A regular hexagon requires

$$g_{\text{gap}} = s \iff p = \frac{\sqrt{3}s}{2}. \quad (\text{M15})$$

For the current shell, $g_{\text{gap}} = s/\sqrt{2}$, so its top measurement hexagon is not regular. Its vertex rays have angles $\alpha_i \pm \gamma$, where $\alpha_i = \arg n_i$ and

$$\gamma = \arctan \frac{s}{2p} = \arctan(\sqrt{3}(\sqrt{2}-1)) \simeq 35.657130^\circ. \quad (\text{M16})$$

Only at regularity is $\gamma = 30^\circ$. The current endpoint rays must not be replaced by the regular hexagon's tangent-axis rays. The remembered “three additional directions” could refer to opposite seam rays, opposite regular-hexagon rays, or tangent versus normal lines. Without a marked source, the intended choice remains open. [TL4 “Top projection and the horizontal direction sets,” (14)–(15).]

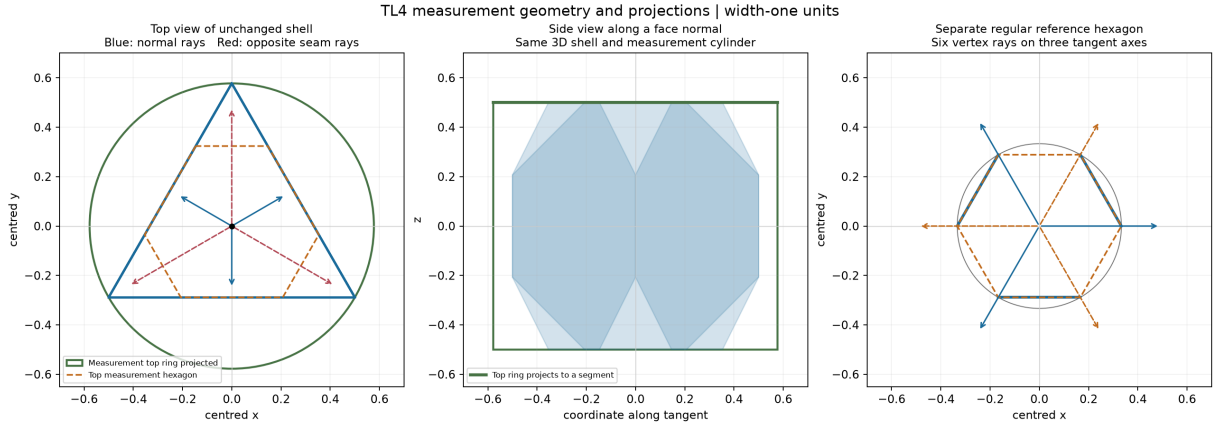


Figure 2: Retained TL4 static views in width-one coordinates: actual shell projections and a separate regular reference hexagon. Surface edges, measurement chords and reference geometry are distinguished by the original labels. Source: TL4, copied byte-for-byte; geometric diagram, not a dynamical simulation. Source hash in the supplement inventory.

Left: triangle perimeter, alternating top measurement chain, and projected cylinder ring. Centre: side projection within the same measurement cylinder. Right: the distinct regular side-1/3 reference hexagon. All three panels use the original width-one unit; the right panel is not a replacement for the welded shell.

5 Normalization changes versus changes of construction

The documented old folded export used octagon side $s_{\text{old}} = 1$, hence width $W_{\text{old}} = 1 + \sqrt{2}$. The current export uses width one. The recovered relation, for corresponding coordinates about the same original origin, is

$$x_{\text{current}} = (\sqrt{2} - 1)x_{\text{old}}. \quad (\text{M17})$$

The symmetry centres obey the same scaling. Lengths scale by $\sqrt{2}-1$, areas by $3-2\sqrt{2}$; incidence and angles do not change. TL4 compared all 18 archived vertices. This conversion belongs to those documented folded exports, not arbitrary historical displays. [TL4 “Two documented scales,” (1).]

Quantity	Current width one	Old side one
Octagon side	$\sqrt{2} - 1$	1
Width/full height	1	$1 + \sqrt{2}$
Octagon apothem	$1/2$	$(1 + \sqrt{2})/2$

Quantity	Current width one	Old side one
Individual octagon circumradius	$\sqrt{1 - \sqrt{2}/2}$	$\sqrt{4 + 2\sqrt{2}}/2$
Full-shell minimum cylinder radius	$1/\sqrt{3}$	$(1 + \sqrt{2})/\sqrt{3}$

The separate fixed-face-centre matching operation holds $p_0 = a_0/\sqrt{3}$ fixed and shrinks each already placed face about its own centre. Its new side is $\lambda_{\text{shrink}} s_0$. Setting the connector equal to that side gives

$$\sqrt{3}p_0 - \lambda_{\text{shrink}} s_0/2 = \lambda_{\text{shrink}} s_0, \quad \boxed{\lambda_{\text{shrink}} = \frac{1 + \sqrt{2}}{3}}. \quad (\text{M18})$$

For initial width one, the resulting reference faces have side and connector $1/3$, half-height $(1 + \sqrt{2})/6$, top hexagon circumradius $1/3$, and whole-frame cylinder radius $\sqrt{6 + 2\sqrt{2}}/6$. The operation loses the welded seams. It is not a change of units and does not describe the current shell.

An alternative retains the face size and translates each centre outwards to $p_* = \sqrt{3}s/2$. Its whole-frame cylinder radius is $\sqrt{(5 - 3\sqrt{2})/2}$, half-height $1/2$. The two regular alternatives are similar to each other by (M18), but not to the original folded arrangement. A global similarity preserves the original ratio $g_{\text{gap}}/s = 1/\sqrt{2}$, so it cannot regularize that hexagon. [TL4 “The hexagon matching calculation,” (4)–(6); inherited Paper D construction.]

The general reference scaffold uses $p = (s + 2g_{\text{gap}})/(2\sqrt{3})$, $a = (1 + \sqrt{2})s/2$, and a different planar octagon-centre radius $L = p + a$. Its Paper-C member is rigidly registered by

$$M(x) = R_z(\pi/6)x + (0, a/\sqrt{3}, 0), \quad p = a/\sqrt{3}. \quad (\text{M19})$$

At $s = \sqrt{2} - 1$, no further scaling is needed. Reference indices 0, 1, 2 map to panels P_3, P_1, P_2 . Complete planar reference octagons and generic family members remain different point sets. Their radial bounds are $\sqrt{(p + 2a)^2 + s^2}/4$ for complete planar frames and $\sqrt{p^2 + a^2}$ for complete vertical frames; the small hexagon only requires $\sqrt{p^2 + s^2}/4$. [TL4 reference registration (20) and footprint tables.]

Unresolved provenance. Approximately 0.544 was not recovered as a radius or shrink normalization. The documented individual-octagon circumradius $0.541196100\dots$ and full-shell cylinder radius $0.577350269\dots$ measure different extrema about different centres. Numerical resemblance supplies no identification. Historical torus display values $L = 2, R_{\text{tube}} = 0.6$ instead give an envelope cylinder of radius 2.6 and half-height 0.6, with the torus’s own highest circle at radius 2. No source fixes that display unit to the current width; their aspect ratios also prevent one similarity from matching both minimum-cylinder dimensions. [TL4 normalization and display tables.]

6 The lens-to-state interface: shape without absolute scale

The retained equal-radius planar lens has parent disks centred at $(\pm d/2, 0)$, with $r > 0, 0 \leq d \leq 2r$. Its filled intersection is

$$\mathcal{L}_{r,d} = \{(u, w) : (|u| + d/2)^2 + w^2 \leq r^2\}. \quad (\text{M20})$$

Set

$$q = \frac{d}{r} \in [0, 2], \quad \theta = \arccos(q/2), \quad I(\theta) = \frac{2\theta - \sin 2\theta}{\pi}, \quad G = \sqrt{I}. \quad (\text{M21})$$

The area is $r^2(2\theta - \sin 2\theta)$, so I is area divided by parent-disk area. The current named response `lens_area_norm_v1` adopts its square root as an amplitude gain. That adoption is not derived from a material aperture law.

Fix the incident vector $\xi \in \mathbb{C}^2$, a named unit eigenbra $\chi^\dagger$, and a finite preparation-transfer count $n \geq 0$. Writing $n = 3m + j$, $j \in \{0, 1, 2\}$, the existing map is

$$\boxed{\Omega_0 = G(\theta)v}, \quad v = (\chi^\dagger \xi) b_\chi^n h_n f_j, \quad h_n = D_s^m(1, q_s, q_s^2 c_s)_j, \quad (\text{M22})$$

where $\|f_j\| = 1$, $|b_\chi| = 1$,

$$q_s = e^{-.423}, \quad r_s = e^{.577}, \quad c_s = .382, \quad D_s = r_s q_s^2 c_s^2 = e^{-.269} (.382)^2 \in (0, 1). \quad (\text{M23})$$

These decimal constants are the retained exact mathematical source constants. Their floating evaluation has its own contract. The count n is preparation, not the later nonlinear recurrence count; r_s is not a lens radius. No normalization of Ω_0 is inserted.

To make the factorization explicit, put $\zeta_3 = e^{2\pi i/3}$ and $f_j = (1, \zeta_3^{-j}, \zeta_3^{-2j})/\sqrt{3}$. The fixed spatial transfer $A_s = R_s Z_s C_s$ acts by $A_s f_0 = q_s f_1$, $A_s f_1 = q_s c_s f_2$, $A_s f_2 = r_s c_s f_0$. Each complete three-step cycle therefore multiplies by D_s ; the remaining zero, one or two steps give the three factors in (M22). The full fixed transfer is $U_s = B_s \otimes A_s$, with helicity-major tensor order. Applying the eigenbra to $U_s^n[G(\xi \otimes f_0)]$ uses $\chi^\dagger B_s^n = b_\chi^n \chi^\dagger$ and yields (M22). The named branches are `negative_imag` and `positive_imag`; their normalized eigenbra is fixed as part of the data. This is the existing SRG preparation, not the nonlinear recurrence. [TL1 §1; native `srg.py`, `fixed_november_srg`, `fourier_basis` and `helicity_mode`.]

Theorem 6.1 (Fixed-data fibres). *Because $q(\alpha r, \alpha d) = q(r, d)$,*

$$\Omega_0(\alpha r, \alpha d) = \Omega_0(r, d), \quad \alpha > 0. \quad (\text{M24})$$

Moreover $I'(\theta) = 4 \sin^2 \theta / \pi$. For any two distinct angles in $[0, \pi/2]$, the integral of this derivative between them is positive. Thus I and G are strictly increasing, including the endpoint where the derivative vanishes. They map onto $[0, 1]$.

If $v \neq 0$, attainable outputs are exactly $w = tv$, $0 \leq t \leq 1$. The noiseless real scalar is $t = v^\dagger w / \|v\|^2 = \|w\| / \|v\|$. Hence $\theta = I^{-1}(t^2)$ and $q = 2 \cos \theta$ are unique. For that output the entire fibre is

$$\{(r, d) : r > 0, d = qr\} = \{(\alpha r_0, \alpha d_0) : \alpha > 0\}. \quad (\text{M25})$$

An output outside the segment has empty fibre. At $t = 1$, $q = 0$: all radii with $d = 0$ coincide. Here separation is known to be zero although radius is arbitrary. At $t = 0$, $q = 2$: the fibre is every tangency pair $(r, 2r)$. Otherwise neither r nor d is individually recovered.

Zero-output classification. Every finite h_n is positive, all Fourier vectors are nonzero, and neither named eigenvalue vanishes. Explicitly $b_\pm = \tau \pm i\sqrt{1 - \tau^2}$, with $\tau = \cos(2\pi \cdot .244) \cos(\pi \cdot .244)$ and $|\tau| < 1$. Therefore

$$\Omega_0 = 0 \iff d = 2r \text{ or } \chi^\dagger \xi = 0. \quad (\text{M26})$$

Nonzero incidence does not guarantee nonzero extraction: for $\chi = (a, b)$, the nonzero vector $(-\bar{b}, \bar{a})$ is orthogonal to χ . If $v = 0$, the **whole** lens domain is one zero fibre and all nonzero-output fibres are empty. No branch/count-specific exact zero exists at finite n beyond these routes. Underflow, precision guards, and cancellation in binary64 are not additional exact mathematical fibres. [TL1 §§2–7; native boundary/SRG definitions.]

The interface forgets absolute common scale even in exact arithmetic. It also has no spatial orientation input. A proposed top-layer radius which matters independently cannot be identified through this initializer alone. This is an **information-interface result**, not a physical no-go theorem and not a reason to add a radius channel without adoption.

7 Conditioning depends on the variable and error model

Assume that v is fixed, known and nonzero. Write $t = G$, $Q(t) = q$, $\Theta(t) = \theta$, and $\delta(t) = 2 - Q(t)$. On the interior, differentiation gives

$$\frac{dt}{d\theta} = \frac{2 \sin^2 \theta}{\pi t}, \quad \frac{dt}{dq} = -\frac{\sin \theta}{\pi t}, \quad \frac{dq}{dt} = -\frac{\pi t}{\sin \theta}, \quad \frac{d\theta}{dt} = \frac{\pi t}{2 \sin^2 \theta}. \quad (\text{M27})$$

These are not endpoint substitution formulas. At coincidence $q = 0, t = 1$, the respective last three derivatives have finite one-sided limits $-1/\pi, -\pi, \pi/2$. At tangency their apparent 0/0 factors require asymptotics.

7.1 Tangency asymptotics

For $\delta = 2 - q \rightarrow 0^+$, the half-angle identity gives

$$\theta = 2 \arcsin(\sqrt{\delta}/2) = \sqrt{\delta}(1 + \delta/24 + 3\delta^2/640 + O(\delta^3)).$$

Since $dI/d\delta = \sqrt{4\delta - \delta^2}/\pi$, expanding and integrating from zero gives

$$I = \frac{4}{3\pi} \delta^{3/2} \left(1 - \frac{3\delta}{40} - \frac{3\delta^2}{896} + O(\delta^3) \right), \quad t = \frac{2}{\sqrt{3\pi}} \delta^{3/4} (1 - 3\delta/80 + O(\delta^2)). \quad (\text{M28})$$

Define $C_\theta = (3\pi/4)^{1/3}$, $C_\delta = C_\theta^2$. Inversion in the fractional-power coordinate produces

$$\delta(t) = C_\delta t^{4/3} + \frac{C_\delta^2}{20} t^{8/3} + O(t^4), \quad \Theta(t) = C_\theta t^{2/3} \left(1 + \frac{C_\delta}{15} t^{4/3} + O(t^{8/3}) \right). \quad (\text{M29})$$

Thus Q and δ are C^1 at tangency, with derivative zero; their first derivatives have local Hölder exponent $1/3$, and second derivatives diverge. Angle recovery is optimally Hölder $2/3$, not Lipschitz or differentiable with finite derivative there. The forward gain is Hölder $3/4$ as a function of q . These exponents follow from the series, not fitted data. [TL2 §3, (5)–(12).]

Exact sharp bound. On the whole gain interval,

$$\boxed{|Q(t_1) - Q(t_2)| \leq \pi |t_1 - t_2|}. \quad (\text{M30})$$

Indeed

$$\frac{d}{d\theta} \frac{I(\theta)}{\sin^2 \theta} = \frac{4(\sin \theta - \theta \cos \theta)}{\pi \sin^3 \theta} \geq 0.$$

The numerator starts at zero and has derivative $\theta \sin \theta \geq 0$. The ratio reaches 1 at $\pi/2$, so $t \leq \sin \theta$ and $|Q'| \leq \pi$. Integrating proves (M30); the slope at $t = 1$ proves sharpness. The same bound holds for δ . Absolute ratio recovery from amplitude is therefore not singular at tangency. Relative condition numbers refer to **relative amplitude error**, $\kappa_f = |tf'(t)/f(t)|$, when the denominator is nonzero:

$$\kappa_\theta = \frac{2\theta - \sin 2\theta}{2\theta \sin^2 \theta}, \quad \kappa_q = \frac{2\theta}{\sin 2\theta} - 1, \quad \kappa_\delta = \frac{2\theta - \sin 2\theta}{2(1 - \cos \theta) \sin \theta}. \quad (\text{M31})$$

Their tangency limits are $2/3, 0, 4/3$; their coincidence limits are $1, +\infty, \pi/2$. Relative errors of an exactly zero angle, gap, or ratio are undefined. If the datum instead has fixed **additive area error**, then $\delta(I) \sim C_\delta I^{2/3}$, $\Theta(I) \sim C_\theta I^{1/3}$, and $|dq/dI| = \pi/(2 \sin \theta) \rightarrow \infty$. This is a different inverse problem. Squaring an amplitude estimate induces area error $2t e_t + e_t^2$, not an independent fixed area-error budget. [TL2 §4, (13)–(16).]

7.2 Arbitrary complex output error

Let $\hat{w} = tv + e$, $\|e\| \leq \varepsilon_{\text{out}}$, and $V_0 = \|v\| > 0$. Minimizing $\|\hat{w} - sv\|^2$ over real s gives

$$\hat{t} = \frac{\text{Re}(v^\dagger \hat{w})}{V_0^2}, \quad |\hat{t} - t| \leq \frac{\|e\|}{V_0} \leq E_{\text{out}} := \frac{\varepsilon_{\text{out}}}{V_0}. \quad (\text{M32})$$

The normal equation follows by differentiating a strictly convex quadratic in s ; the bound is Cauchy–Schwarz, sharp for real parallel error. If $e = (a_e + ib_e)v + e_H$, $v^\dagger e_H = 0$, then $\hat{t} - t = a_e$. Both quadrature error $ib_e v$ and Hermitian-orthogonal error leave that scalar estimate unchanged, although they contribute residual. Estimating t by $\|\hat{w}\|/V_0$ would not have this property.

Clipping the estimate to $[0, 1]$ is nonexpansive relative to the true attainable t . Thus

$$|Q(\tilde{t}) - Q(t)| \leq \min(2, \pi E_{\text{out}}). \quad (\text{M33})$$

This is estimator analysis, not an initializer change. If $r_\perp = \hat{w} - \hat{t}v$, the exact feasible gain interval is

$$\left[\hat{t} - \frac{\sqrt{\varepsilon_{\text{out}}^2 - \|r_\perp\|^2}}{V_0}, \hat{t} + \frac{\sqrt{\varepsilon_{\text{out}}^2 - \|r_\perp\|^2}}{V_0} \right] \cap [0, 1], \quad (\text{M34})$$

provided the square root is real and the intersection nonempty. Otherwise the observation is incompatible with the error model. Applying decreasing Q reverses endpoint order.

In the resolved regime $E_{\text{out}} \ll t \ll 1$, local errors scale as $(4/3)C_\delta t^{1/3}E_{\text{out}}$ for the ratio/gap and $(2/3)C_\theta t^{-1/3}E_{\text{out}}$ for the angle. These linearized estimates cannot be extrapolated to fixed error at $t = 0$. If $t \leq E_{\text{out}}$, an admissible error can erase the output entirely. At true tangency the exact worst-case errors are $\delta(\min(E_{\text{out}}, 1))$ and $\Theta(\min(E_{\text{out}}, 1))$, hence asymptotically $C_\delta E_{\text{out}}^{4/3}$ and $C_\theta E_{\text{out}}^{2/3}$. A small gap can be unresolved relatively despite controlled absolute ratio error. [TL2 §§5–6, (17)–(23).]

7.3 Extraction attenuation is a separate condition number

Equation (M22) gives $V_0 = |\chi^\dagger \xi| h_n$. Put $p_0^{(s)} = 1, p_1^{(s)} = q_s, p_2^{(s)} = q_s^2 c_s, \rho_s = D_s^{1/3}$. Then

$$h_n = p_j^{(s)} \rho_s^{-j} \rho_s^n > 0, \quad h_{n+3} = D_s h_n, \quad E_{\text{out}} = \frac{\varepsilon_{\text{out}}}{|\chi^\dagger \xi| h_n}. \quad (\text{M35})$$

Here $D_s \simeq 0.1115068404$, $\rho_s \simeq 0.4813199210$. Each three extra transfers multiplies fixed absolute-error amplification by $D_s^{-1} \simeq 8.96806$; the per-count asymptotic factor $\rho_s^{-1} \simeq 2.07762$ is not the ratio at every individual step. Every finite count remains exactly nonzero. This is attenuation, not exact information loss.

Likewise $|\chi^\dagger \xi| \rightarrow 0^+$ makes absolute-error amplification unbounded while preserving noiseless shape injectivity. At overlap exactly zero the whole-domain collapse (M26) occurs. With a purely relative output budget $\|e\| \leq \varepsilon_{\text{rel}} t V_0$, the extraction norm cancels and the local relative condition numbers apply. No stochastic assumption is used. No improvement in precision or gain can recover absolute r , since different radii already have identical **noiseless** data. [TL2 §§7–8, (24)–(27).]

8 Changing a view versus orienting the eye

An orthonormal screen $E = [a_0 \ b_0]$, with sight direction $c_0 = a_0 \times b_0$, gives coordinates $E^T(x - o)$ and ambient projection $I - c_0 c_0^T$. For two points,

$$\|E^T(x - y)\|^2 = \|x - y\|^2 - |c_0 \cdot (x - y)|^2. \quad (\text{M36})$$

Depth and projected lengths change with the view; the object's distances do not. Top and side screens were used above. A face-on lens embedding is $x = o + ua_0 + wb_0$, with (M20) in (u, w) .

To illustrate a quarter-turn without adopting a spin axis, conditionally take $Qa_0 = a_0$, $Qb_0 = c_0$, $Qc_0 = -b_0$. An active rotation changes the eye to $o + ua_0 + wc_0$ relative to the fixed scaffold. In its old screen it appears as $(u, 0)$. A view change alone to screen (a_0, c_0) makes the original fixed lens appear as $(u, 0)$ as well. If object and screen rotate together,

$$(QE)^T Q(x - o) = E^T(x - o). \quad (\text{M37})$$

The image is unchanged although the eye's relation to the fixed shell may have changed. A passive coordinate change applied to every object changes neither relation. These examples prove why a changed silhouette does not determine an active rotation. [TL4 (16)–(19).]

For a generic lens $0 < q < 2$, an initial placement needs a plane **and** its in-plane centreline: a normal alone leaves one angle free. A marked right-handed frame $Q = [a_0 \ b_0 \ c_0] \in SO(3)$ supplies three orientation degrees of freedom. For only the unlabeled lens set, half-turns about each frame axis preserve it, giving $SO(3)/D_2$, with D_2 the four-element group of those proper rotations. At $q = 0$, a disk needs only its plane, represented by an unoriented normal in $\mathbb{RP}^2$. At $q = 2$, the overlap point has no orientation; the retained parent pair would still have one.

Polar coordinates and rotation axes also transform differently under reflection. For an orthogonal shell symmetry g ,

$$gQ_a(\alpha)g^{-1} = Q_{\det(g)ga}(\alpha). \quad (\text{M38})$$

Thus an axis with signed rotation is axial, whereas an ordinary displacement is polar. The existing D_{3h} symmetry relates equivalent registrations but supplies no continuous spin evolution. [TL3 §3; TL4 “Vesica eye orientation without dynamics.”]

9 What the conditional host calculations establish

TL3 asked a different, conditional question: after selecting a 3D lift of the lens and registering it at o , what scale encloses the unchanged panels? Its exact answers are retained in Appendix A.

Revolving the lens around either planar symmetry axis gives two different convex solids. Their vertical versions have uppermost **points**. A circular path sweep can instead have a highest circle, but needs an extra path radius and a profile-orientation choice. An open-hole torus can contain the panels while excluding the centre, so it fails the stronger TL4 measurement requirement.

A rigid quarter-turn of a planar lens stays planar; it cannot itself enclose a panel set of affine rank three. No viewpoint change repairs that dimensional mismatch. Conversely a conditional containment bound supplies neither a material attachment nor a radius inferred from Ω_0 . Its minimum-scale rule is a design condition for the chosen host and registration. [TL3 §§3–9.]

10 Conclusions and remaining choices

The clarified measurement object is a fixed centred vertical cylinder with the half-height convention (M8); its minimum dimensions and top ring follow from actual panel extrema. The top-view triangle, alternating top measurement hexagon, and separately regularized reference hexagon have different definitions. Opposite seam rays add directed rays on existing normal axes; tangent axes are another already determined set.

The adopted initializer retains shape ratio at fixed known nonzero extraction, with precisely the scaling fibres (M25). Absolute radius is absent, not numerically hidden. Ratio inversion is sharply Lipschitz under amplitude error; angle inversion and an additive-area-error problem have different endpoint regularity. Attenuation and near-orthogonal extraction affect practical resolution without changing exact fibres until extraction is exactly zero.

What remains unspecified is the eye object to retain, its absolute scale and initial frame relative to the scaffold, labels on its parents or normal, and any independent orientation evolution or observable. The approximate 0.544 provenance and the exact historical “additional directions” remain unresolved. A future attachment can be tested against the established fibres only after that attachment has been independently defined. No follow-on calculation begins here.

A Retained alternative containment formulas

These are **conditional definitions and exact containment results**, not adopted material hosts. Points are centred at o . To distinguish host profile scale from cylinder radius, write TL3's parent scale as R_L , its shape as $0 \leq q < 2$, and

$$c_q = 1 - q^2/4, \quad a_q = 1 - q/2, \quad \Phi_q(M, K) = \frac{qM + \sqrt{q^2 M^2 + 4c_q K}}{2c_q}. \quad (\text{MA1})$$

This is the nonnegative root of $c_q R_L^2 - qM R_L - K = 0$. For a chosen unit axis e , write $z_e = p' \cdot e$, $\rho_e^2 = \|p'\|^2 - z_e^2$.

The two convex revolutions. Rotating (M20) around its centreline gives an intersection of balls:

$$K_C = \{p' : \rho_e^2 + (|z_e| + qR_L/2)^2 \leq R_L^2\}. \quad (\text{MA2})$$

Rotation around the chord instead gives

$$K_H = \{p' : (\rho_e + qR_L/2)^2 + z_e^2 \leq R_L^2\}. \quad (\text{MA3})$$

The inequalities are respectively $\|p'\|^2 + qR_L|z_e| \leq c_q R_L^2$ and $\|p'\|^2 + qR_L \rho_e \leq c_q R_L^2$, proving convexity. Every panel is the convex hull of its vertices, so vertex containment is sufficient and necessary. All centred vertices p_j have norm B , giving

$$R_{L,C}^{\min} = \Phi_q(\max_j |p_j \cdot e|, B^2), \quad R_{L,H}^{\min} = \Phi_q(\max_j \sqrt{B^2 - (p_j \cdot e)^2}, B^2). \quad (\text{MA4})$$

Maximizing vertices attain contact, proving sharpness. For axes e_z, n_i, t_i , the first maxima are $1/2, 1/\sqrt{3}, 1/2$; the second are $1/\sqrt{3}, \sqrt{3}/8, B$. At $q = 0$ both hosts are the sharp sphere of radius B . For $q < 2$ their boundaries are topological spheres, not tori. For vertical axes their highest points occur at $z = R_L a_q$ and $z = R_L \sqrt{c_q}$, respectively. Interior parallels are circles, but no positive-radius highest ring arises. At $q = 2$ the overlap lifts collapse and cannot enclose S . [TL3 (10)–(14).]

Capped extrusion. A frame (a_0, b_0, c_0) and half-depth H_L define

$$(|p' \cdot a_0| + qR_L/2)^2 + (p' \cdot b_0)^2 \leq R_L^2, \quad |p' \cdot c_0| \leq H_L. \quad (\text{MA5})$$

Convexity gives the exact vertex test

$$H_L \geq \max_j |p_j \cdot c_0|, \quad R_L \geq \max_j \Phi_q(|p_j \cdot a_0|, (p_j \cdot a_0)^2 + (p_j \cdot b_0)^2). \quad (\text{MA6})$$

The caps are part of the boundary; sweeping only lens arcs leaves an open side wall. Depth is additional data. [TL3 (15)–(17).]

Circular sweeps. With path radius $L \geq 0$, vertical axis e_z , and $\rho \geq 0$, TL3 retains two orientations:

$$K_A : (\rho - L)^2 + (|z| + qR_L/2)^2 \leq R_L^2, \quad K_R : (|\rho - L| + qR_L/2)^2 + z^2 \leq R_L^2. \quad (\text{MA7})$$

The axial and radial parent-centre choices are not interchangeable registrations. Define three actual panel test types

$$(\rho_0, Z_0) = (p, 1/2), \quad (\rho_1, Z_1) = (v_6, 1/2), \quad (\rho_2, Z_2) = (1/\sqrt{3}, s/2), \quad D_i^2 = (\rho_i - L)^2 + Z_i^2.$$

Then full-panel containment is exactly

$$R_L \geq \max_i \Phi_q(Z_i, D_i^2) \quad (K_A), \quad R_L \geq \max_i \Phi_q(|\rho_i - L|, D_i^2) \quad (K_R). \quad (\text{MA8})$$

The extra test $(p, 1/2)$ is a top-edge midpoint. These hosts can be nonconvex, so the earlier vertex argument would be invalid. TL3's proof first maximizes over $|z|$ in (M5). On $0 \leq u \leq s/2$, height is constant and each expression is convex in ρ , giving its endpoints. On the chamfer, $\rho = \sqrt{p^2 + u^2}$, $z = 1/\sqrt{2} - u$. The axial expression has second derivative $4 - 2Lp^2/(p^2 + u^2)^{3/2}$, increasing with u , and initial first derivative $s - 1 - Ls/v_6 - qR_L < 0$; it has no interior maximum. On each radial-centres piece use $L' = L - \sigma qR_L/2$, $\sigma = \text{sign}(\rho - L)$. For $L' < 0$ it is strictly convex; otherwise the preceding derivative argument applies. At $\rho = L$ the first derivative jumps upwards. Thus the three displayed endpoint types suffice. Solving their quadratics proves (MA8). [TL3 §6.2, (22)–(24).]

For $L > 0$ the highest sets are circles of radius L , at $z = R_L a_q$ for K_A , or $R_L \sqrt{c_q}$ for K_R . An open hole requires $L > R_L \sqrt{c_q}$ or $L > R_L a_q$, respectively. There the solid is a disk times a circle and its boundary an embedded torus, possibly creased. At equality the inner circle pinches to an axial point. After the profile crosses the axis the boundary of the swept solid is sphere-like; a retained self-overlapping parametrization is a different object.

For K_A , an enclosing open-hole torus exists exactly when

$$0 \leq q < 1, \quad L > \frac{1/3}{2p - q/(2\sqrt{c_q})}, \quad (\text{MA9})$$

with $R_{L,A}^{\min} \leq R_L < L/\sqrt{c_q}$. For K_R , the exact condition is $\max_i \Phi_q(|\rho_i - L|, D_i^2) < L/a_q$, feasible for every $q < 2$, for example with $L > 1/\sqrt{3}$. These are TL3's fixed-registration results, not optimizations over all hosts. Every open hole excludes o , disqualifying it as TL4's centre-containing measurement domain. [TL3 (25)–(26) and topology discussion.]

B Historical source map without a new census

TL0's 24 records are sixteen specified constructions and eight incomplete proposals, not 24 inequivalent circles. This compact map retains every identifier; primary historical locators remain in TL0 §§2–4.

TL0 records	Retained object type and limitation
H01–H03	DMQPF scalar perimeter/profile; equal parent circles; overlap lens. Scalar corrections are not an embedded shell law.
H04–H09	VPQW toroidal proposal; three-octagon placement; E8 orbit claim; D24 orientation frame; rosette; Möbius-labelled recurrence. H04, H06, H08, H09 lack complete relevant geometry.
H10–H12	Passive discrete clock, scalar/macro/chiral readouts, cylindrical history curve. None supplies a fixed upper circle.
H13–H16	History-dependent torus chart, state-dependent three-anchor chart, fixed torus wireframe, RSB halo parallels. Their radii and roles differ.
H17–H20	Polar occupancy histogram, planar limaçon, angular-gap predicate, prescribed time-dependent quiver. These are not one common torus.

TL0 records	Retained object type and limitation
H21–H24	Outer long-return schematic, meta-shell feedback proposal, nominal seed-manifold union, larger circular/Vesica-host intent. All four have incomplete geometric/attachment definitions.

H24 is the strongest retained upstream **intent**; H02/H03 give an exact profile; H15 gives a display precedent for a highest torus parallel. No recovered equation identifies all three. The later TL4 definition selects a measurement meaning without rewriting this history. The scripts `tri.py` and `center3.py` cited by the historical placement were not recovered in the inspected source set. TL3 likewise recovered no exact transformation identified as the remembered de Broglie/Vesica flip. These are bounded source-recovery limitations, not universal absence claims.

C Notation, claims, evidence, and source locations

This text	Source notation	Distinction
p	TL3 h ; TL4 p	Face-centre distance, not a dynamical step
R, H	TL4 R, H	Cylinder radius and half-height
R_L, H_L	TL3 R, H	Conditional lens-host scale/depth
v_6	TL3 v ; TL4 ρ_6	High-vertex horizontal radius; v in §6 is the extraction vector
C_θ, C_δ	TL2 B, A	Tangency constants
$\varepsilon_{\text{out}}, E_{\text{out}}, V_0$	TL2 η, E, V	Output error, gain error, extraction norm
$\lambda_{\text{shrink}}, g_{\text{gap}}$	TL4 shrink λ , gap g	Neither is a recurrence coefficient
$\rho_s, p_j^{(s)}$	TL2 ρ, p_j	Transfer attenuation, not spatial radius/point

Central claim/ equation	Exact source locator	Evidence type
Shell coordinates and bounds: M1–M7	TL3 §2 (1)–(3); native geometry/frame definitions	Inherited definition; derived identity
Cylinder, top ring, enclosure: M8–M11	TL4, “Measurement cylinder,” (7)–(11)	Owner-guided definition; exact theorem
Projections/directions: M12–M16	TL4 “Top projection,” (12)–(15); “Viewpoint,” (16)–(18)	Derived identities with full-set proofs
Conversion, shrink, registration: M17–M19	TL4 normalization (1)–(6), registration (20)	Recovered source fact; inherited Paper C/D geometry
Lens fibres and zero routes: M20–M26	TL1 §§2–7; TL3 (6)–(9)	Exact theorem; information-interface result
Conditioning/error/gain: M27–M35	TL2 §§2–8 (2)–(27)	Derived identities; asymptotics; error theorem
Views/orientation: M36–M38	TL4 “Viewpoint” and “Vesica eye orientation,” (16)–(21); TL3 §3 (5)	Kinematic definition; exact identities
Host appendix MA1–MA9	TL3 §§5–6 (10)–(26)	Conditional construction; exact containment
Historical unresolved identity	TL0 §§2–9; TL4 remembered-decimal discussion	Source-limited recovery; open questions

Retained checks are **not rerun here** and are not independent-theorem counts:

Checkpoint	Recorded passing count	Categories as stored
TL0	63/63	46 source facts; 9 derived identities; 8 numerical checks
TL1	24/24	7 derived identities; 8 numerical groups; 9 source/preservation checks
TL2	29/29	5 derived identities; 6 asymptotic; 1 information-interface; 8 numerical; 9 preservation
TL3	35/35	11 source facts; 18 derived identities; 6 numerical
TL4	45/45	13 source facts; 32 derived identities

The original results retain predicate names, arithmetic and preservation details. TL3’s panel samples support, but do not replace, its endpoint proof. TL4’s archived-vertex check supports the stated conversion, not a conversion for every historical display. This synthesis introduces no new numerical claim or physics interpretation. Public derivatives replace private locators and omit unrelated operational snapshots. Portable commands, original and public hashes, and their tested scope are documented in the joint README and export map.

D Fixed SRG matrices and branch convention

This transcription specifies the adopted preparation in M22–M23 from the pinned native source. Put $\zeta_3 = e^{2\pi i/3}$, $f_j = (1, \zeta_3^{-j}, \zeta_3^{-2j})^T / \sqrt{3}$, $\Pi_0 = f_0 f_0^\dagger$. In the standard channel basis,

$$R_s = q_s I_3 + (r_s - q_s) \Pi_0, \quad C_s = c_s I_3 + (1 - c_s) \Pi_0, \\ Z_s = \text{diag}(1, \zeta_3^{-1}, \zeta_3), \quad A_s = R_s Z_s C_s.$$

Here $q_s = e^{-0.423}$, $r_s = e^{0.577}$, $c_s = 1 - 0.618 = 0.382$. With $\alpha_s = 2\pi(0.244)$, $\beta_s = \pi(0.244)$,

$$B_s = \begin{pmatrix} e^{-i\alpha_s} & 0 \\ 0 & e^{i\alpha_s} \end{pmatrix} \begin{pmatrix} \cos \beta_s & -i \sin \beta_s \\ -i \sin \beta_s & \cos \beta_s \end{pmatrix}, \quad U_s = B_s \otimes A_s.$$

The tensor order is helicity-major $\mathbb{C}^2 \otimes \mathbb{C}^3$. Set $\tau_s = \cos \alpha_s \cos \beta_s$, $b_\pm = \tau_s \pm i\sqrt{1 - \tau_s^2}$. For the named positive_imag or negative_imag branch b , define

$$\Pi_b = \frac{B_s - \bar{b} I_2}{b - \bar{b}}, \quad \chi = \frac{\Pi_b e_p}{\sqrt{(\Pi_b)_{pp}}}.$$

The pivot p maximizes the real diagonal of Π_b , taking the first index on an exact tie; its component is positive real. Since B_s is unitary, this is an orthogonal spectral projector and $\chi^\dagger B_s = b \chi^\dagger$. Input is $G(\xi \otimes f_0)$; extraction is $\chi^\dagger \otimes I_3$. This negative-exponent Fourier basis is the native SRG convention. The companion drift paper’s positive-exponent chiral reference is a different object. Source: pinned `srg.py`, `fourier_basis`, `fixed_november_srg`, `helicity_mode`; copied in the native software supplement with the original license scope.

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Code and evidence availability

The source project is <https://github.com/pzychozen/trioctagon-physics>. Scientific definitions use the unchanged baseline commit `82cab10cbe550f58c43163fb8b05fabdad1b05ae`. The planned publication-package location is `papers/MEASUREMENT_AND_CHIRAL_DRIFT/v0.1/` in that repository. These final files are prepared; repository publication is pending. The shared package contains two separate papers, their sources, figures and supporting evidence.

The complete extracted package provides the local supplement links used here. Its source inventory and public-export map distinguish unchanged copies from labeled public derivatives, retain both raw-original and public-file hashes, and specify exact operational omissions and locator substitutions. Scientific data are preserved; private raw records remain separately archived. The joint README documents bounded build and smoke commands using the pinned native subset and unchanged verifier function bodies. Historical writers must not be executed in-place; the active tools use designated output records. Full historical suites and reviewer companion scripts were not regenerated. Retained evidence, prior review receipts and final packaging checks remain separate.

The software subset retains its original Apache-2.0 license and scope notice. That license is not assigned to manuscripts, research scripts, results or figures. This packaging grants no new redistribution rights and claims no external peer review, journal acceptance or DOI.

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